

ANALYSIS OF RESTAURANT REVENUE

Using Multiple Linear Regression

A Statistical Analysis Report

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Executive Summary

This report analyzed the factors affecting the monthly revenue of restaurants using a dataset of 100 restaurants through Multiple Linear Regression (MLR) modeling. The first model (Model 1) includes all quantitative variables, while the second model (Model 2) expands on the quantitative model to also include qualitative variables. The variable `seating_capacity` was removed from Model 2 because it is highly correlated with `staff_count` and is not statistically significant to monthly revenue. Additionally, Model 3 was constructed to satisfy all regression assumptions and provide statistically significant results for all predictors.

The results in Model 3 indicate that all predictors have meaningful impacts on restaurant revenue. In particular, restaurants in Mall and Downtown areas perform better than those in Suburban locations, while higher customer ratings and increased marketing spending contribute positively to revenue growth.

Several recommendations are proposed based on the results. Restaurant owners are advised to expand their restaurants in Mall and Downtown locations, increase marketing spending, and focus on improving service quality and food consistency to improve customer ratings. These strategies help restaurant owners improve overall business performance and maximize monthly revenue.

2.0 Introduction

In the modern era, restaurant businesses operate in a highly competitive industry, with revenue influenced by many factors such as pricing, location, restaurant type, and more. Understanding how these factors affect monthly revenue is important for restaurant owners to react to and make appropriate decisions.

This study aims to determine the relationship between restaurant characteristics and monthly revenue using a dataset of 100 restaurants. Multiple Linear Regression (MLR) is used to examine several variables simultaneously while holding other variables constant. Issues such as multicollinearity, heteroscedasticity, and normality of residuals are also addressed to ensure the reliability of the results.

This paper first presents the Data Exploration and Methodology section, describing the dataset used in the study and explaining data preparation steps such as the creation of dummy variables. Then, the Model Development and Results section presents model comparisons using statistical measures such as Adjusted R^2 and standard error. Finally, the Business Implications and Recommendations section highlights recommendations that restaurant owners can implement, based on the statistical findings of Model 3.

3.0 Data Exploration and Methodology

The `restaurant_revenue_dataset` is used to conduct this analysis. It contains 100 observations of restaurants with no missing values. `location_type`, `cuisine_type`, `restaurant_size`, `has_parking`, and `has_delivery` are the qualitative variables, while `seating_capacity`, `avg_meal_price`, `years_in_business`,

staff_count, marketing_spend, customer_rating, and monthly_revenue are the quantitative variables within the dataset. monthly_revenue is used as the dependent variable, while the remaining attributes serve as independent variables. The dependent variable (monthly_revenue) is right-skewed, meaning that most restaurants earn lower revenue while only a small number earn high revenue, creating a long tail toward the right side of the chart, as shown in Figure 1.

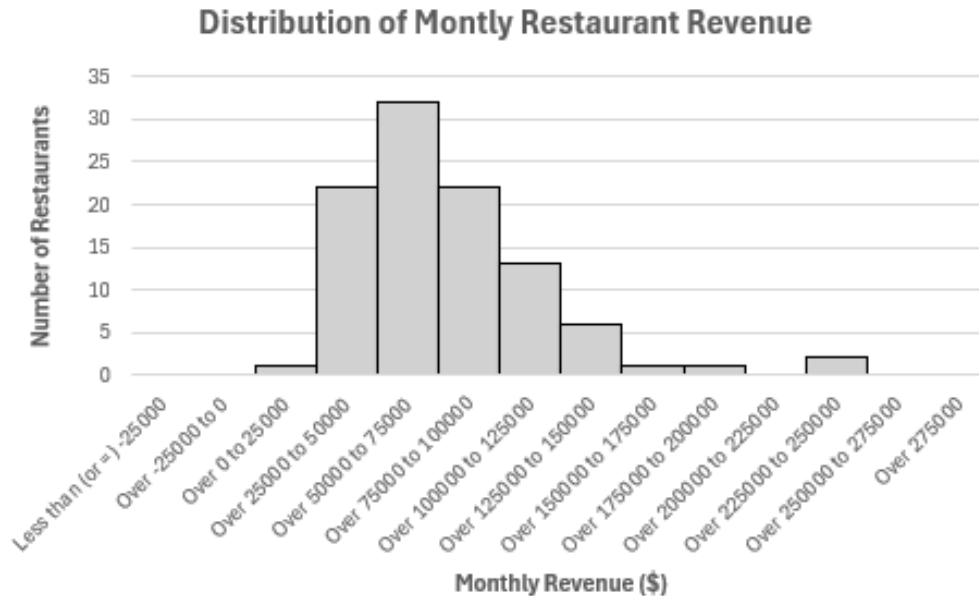


Figure 1: Histogram Distribution of monthly_revenue

Since some of the independent variables in the model, such as cuisine type and location, are qualitative data, they were converted into dummy variables to be included in the regression analysis. Each category was represented by a binary variable coded as 0 or 1. This approach allows qualitative variables to be included in the analysis.

During analysis, a very strong correlation was observed between seating_capacity and staff_count ($r = 0.7776$), as shown in Figure 2. This suggests that restaurants with a larger seating capacity tend to employ more staff. This strong correlation suggests potential multicollinearity, which needs to be considered in further regression analysis.

Correlation Matrix							
	seating_capacity	avg_meal_price	years_in_business	staff_count	marketing_spend	customer_rating	monthly_revenue
seating_capacity	1						
avg_meal_price	0.09993088	1					
years_in_business	-0.040407801	0.003323508	1				
staff_count	0.777624167	0.048505904	-0.036659424	1			
marketing_spend	0.290959152	0.06296475	-0.167261298	0.167802877	1		
customer_rating	0.040224152	-0.00212518	0.022653103	-0.056839945	0.089833506	1	
monthly_revenue	0.287906906	0.214784304	0.074820575	0.379465964	0.249796484	0.235750358	1

Figure 2: Correlation Matrix of Quantitative Variables

This study uses Multiple Linear Regression (MLR) to identify the relationship between independent variables and the monthly revenue of restaurants. MLR is used because it allows more than one

independent variable to be analyzed while controlling for the influence of other variables. Furthermore, MLR allows the identification of which factors are statistically significant with respect to the dependent variable, monthly revenue.

To compare which model performs better, Adjusted R^2 is used to assess how well each model explains the variation while penalizing additional variables, with a higher Adjusted R^2 indicating a better model. Standard Error (SE) is also used to measure how close the predictions are to the actual values, with a smaller SE indicating a better model. Residual diagnostics are also used to compare the models, where a residual plot with a random scatter of residuals indicates that the linearity and homoscedasticity assumptions are satisfied — a good sign for a well-specified model.

4.0 Model Development and Results

4.1 Model 1: Quantitative Variables

In Model 1, only the quantitative variables were used to develop the model. These variables were seating_capacity, avg_meal_price, years_in_business, staff_count, marketing_spend, and customer_rating. After performing the regression test with monthly_revenue as the response variable, the model produced an Adjusted R^2 of 0.2509 and an SE of 34,640.7. The analysis shows that seating_capacity and years_in_business have p-values greater than 0.05, and their confidence intervals include 0. This suggests that these variables are not statistically significant to the monthly revenue of the restaurant. In terms of its residual plot, the graph shows a patterned distribution — the dots spread out more as the fitted value increases, as shown in Figure 3. This indicates heteroscedasticity, meaning the variance of errors is not constant across predictions, and the regression assumption is violated.

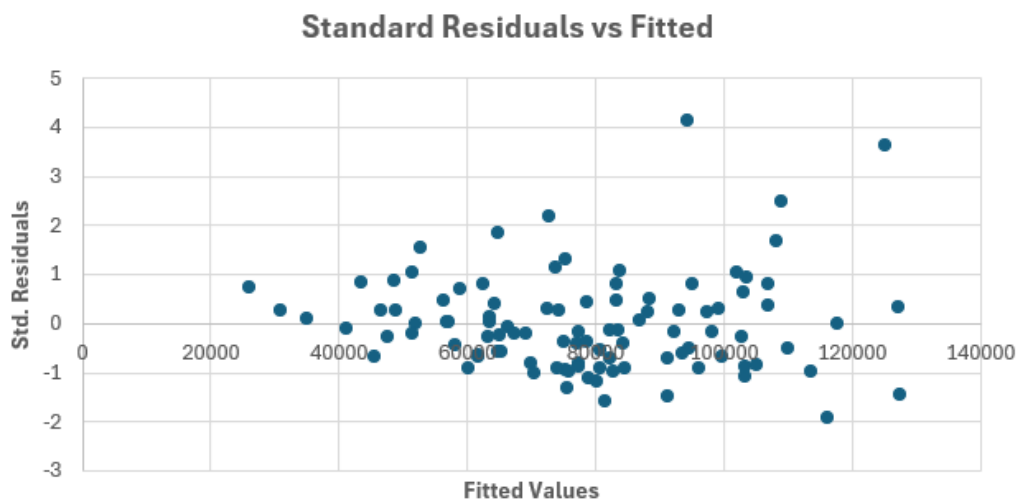


Figure 3: Residual Plot for Model 1

4.2 Model 2: Full Variables

All variables, both quantitative and qualitative, were included in this model. Qualitative variables including `location_type`, `cuisine_type`, `restaurant_size`, `has_parking`, and `has_delivery` were converted into dummy variables and included in the analysis. For each categorical variable, one category was chosen as the reference group to avoid multicollinearity. For instance, for `location_type`, “Suburban” was used as the reference category, and for `cuisine_type`, “Fast_Food” was the reference category. However, Microsoft Excel imposes a limit on the number of independent variables that can be included. Since the model exceeded this limit, `seating_capacity` was ultimately removed. It has a high correlation with `staff_count` ($r = 0.7776$), as shown in Figure 2, which creates a multicollinearity problem. Furthermore, `seating_capacity` is not statistically significant to monthly revenue in Model 1. This further validates that `seating_capacity` is the most suitable candidate for removal in Model 2.

After performing the regression test with `monthly_revenue` as the response variable, the model produced an Adjusted R^2 of 0.8462 and an SE of 15,696.7. The analysis shows that `cuisine_type_Italian` and `has_parking_Yes` have p-values greater than 0.05, and their confidence intervals include 0. This suggests that these variables are not statistically significant to monthly revenue. In terms of its residual plot, the graph shows a patterned distribution similar to that of Model 1, as shown in Figure 4.

Model 2 has a higher Adjusted R^2 compared to Model 1, indicating that Model 2 explains more of the variance in monthly revenue. Additionally, the standard errors of the coefficients in Model 2 are smaller, suggesting more precise estimates. `years_in_business`, which was not significant in Model 1, became statistically significant in Model 2. In conclusion, the addition of dummy variables enhances the regression model, and Model 2 outperforms Model 1.

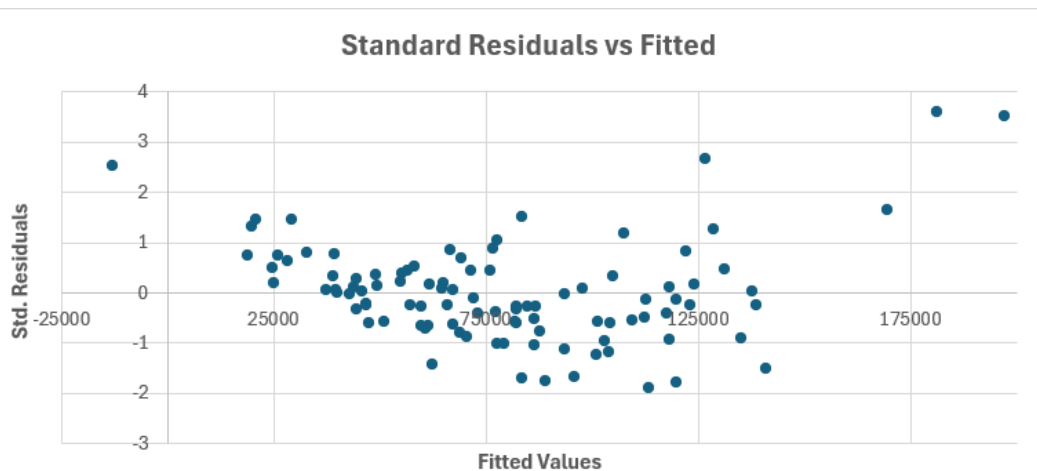


Figure 4: Residual Plot for Model 2

4.3 Model 3: Improved Model

4.3.1 Variable Selection Process and Regression Results

The dependent variable, `monthly_revenue`, was log-transformed to improve the reliability of the results. This transformation was applied so that residuals would be more closely normally distributed, satisfying the assumptions of multiple linear regression, as shown in Figure 5. Variables that were not statistically

significant to the response variable were removed in Model 3 (`cuisine_type_Italian` and `has_parking_Yes`), as the evidence was not strong enough to conclude that these variables influence monthly revenue. `seating_capacity` was also excluded from Model 3 due to its high correlation with `staff_count`.

After performing the regression test with $\ln(\text{monthly_revenue})$ as the response variable, the model produced an Adjusted R^2 of 0.90935 and an SE of 0.1393. The analysis shows no independent variable with a p-value greater than 0.05, and none of the confidence intervals include 0. This suggests that all independent variables are statistically significant, meaning each has a meaningful effect on the response variable.

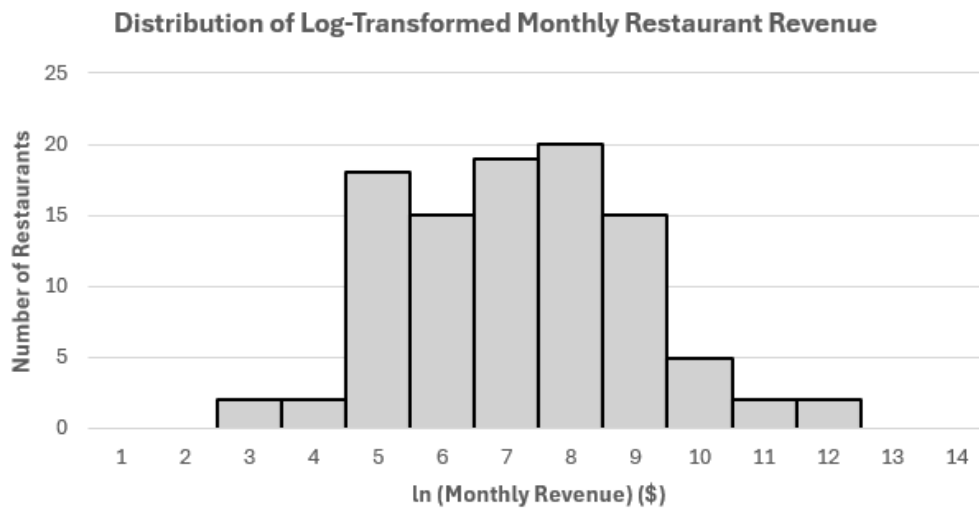


Figure 5: Histogram Distribution of $\ln(\text{monthly_revenue})$

4.3.2 Model 3 Compared with Model 1 and Model 2

Because the dependent variable in Model 3 was log-transformed, the Adjusted R^2 value from this model cannot be directly compared with the previous models. A higher Adjusted R^2 does not necessarily reflect a better model, as these models are not on the same scale.

Residual diagnostics are therefore necessary to compare the models. In Model 3, the residuals are randomly scattered without a clear pattern, indicating that the model captures the relationship between the independent and dependent variables accurately, as shown in Figure 6. Hence, the assumptions of multiple linear regression are satisfied for Model 3. Compared to Model 1 and Model 2, both of which exhibited heteroscedasticity and violated this assumption, Model 3 is shown to be the best model.

Furthermore, two predictors were insignificant in each of Model 1 and Model 2. In the log-transformed model (Model 3), every variable contributes meaningfully to explaining the variation in the dependent variable.

Although the Adjusted R^2 of Model 3 cannot be directly compared to Models 1 and 2 due to the log transformation of the dependent variable, Model 3 is considered the best model overall.

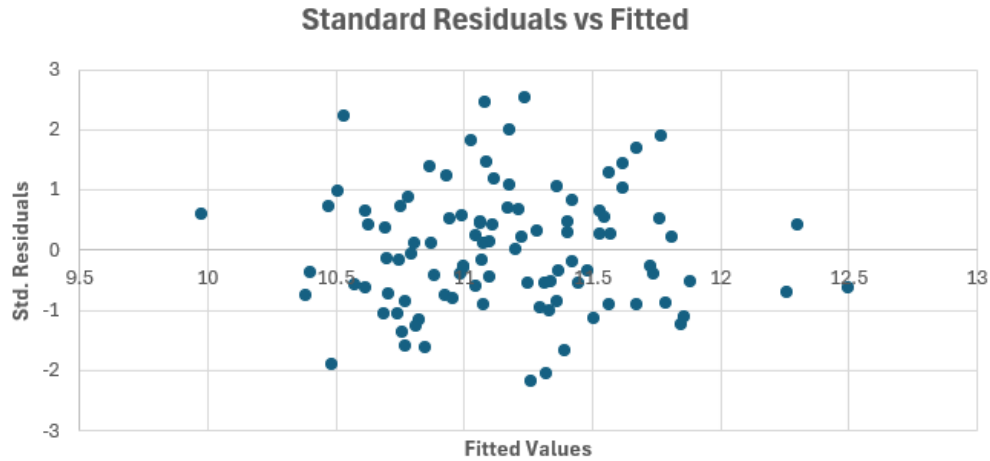


Figure 6: Residual Plot for Model 3

4.4 Model Comparison Table

	Model 1	Model 2	Model 3
Adjusted R ²	0.2509	0.8462	0.90935
Number of Predictors	6	16	14
Insignificant Variables (p > 0.05)	2	2	0
Standard Error	34,640.7	15,696.7	0.1393
Residual Behavior	Heteroscedasticity	Heteroscedasticity	Randomly distributed

5.0 Business Implications and Recommendations

5.1 Economic Interpretation of Key Variables

The results from Model 3 provide a meaningful interpretation of the impact of each variable on monthly revenue. For example, approximately 1.75% of revenue is expected to increase for each one-year increase in `years_in_business`, holding other variables constant, and a 2.13% increase in revenue is expected for each unit increase in `avg_meal_price`, holding other variables constant. For categorical variables, compared to the reference category Suburban, restaurants in Strip Malls are expected to earn approximately 26% less revenue (calculated as $(e^{-0.3012} - 1) \times 100$), those in Malls are expected to earn about 22% more, and those in Downtown areas are expected to earn about 38% more, holding all other variables constant.

5.2 Actionable Business Recommendations

Based on the multiple linear regression results with $\ln(\text{monthly_revenue})$ as the dependent variable, the following recommendations are proposed based on the estimated coefficients.

5.2.1 Recommendation 1: Consider Mall and Downtown Locations for Expansion

Restaurants located in Mall and Downtown areas generate much higher revenue than the reference category, Suburban, with coefficients of $\beta = 0.2019$ for Mall and $\beta = 0.3239$ for Downtown. Hence, restaurant owners planning to open a new restaurant should consider prioritizing Mall and Downtown locations, while existing restaurants located in Strip Malls should consider relocating or implementing a new marketing strategy.

5.2.2 Recommendation 2: Increase Marketing Spending

Marketing spend has a positive and statistically significant relationship with monthly revenue. Although the coefficient is small, a consistent increase in marketing expenditure supports revenue growth over time. Owners should increase their expenditure on marketing, such as digital advertising, and focus on customer loyalty programs.

5.2.3 Recommendation 3: Improve Customer Ratings

Customer ratings show a positive effect on monthly revenue ($\beta = 0.2088$), meaning an approximately 20.9% increase in monthly revenue is associated with a one-point increase in rating, holding other factors constant. Thus, restaurant owners should pay greater attention to service quality and food consistency to improve their customer ratings.

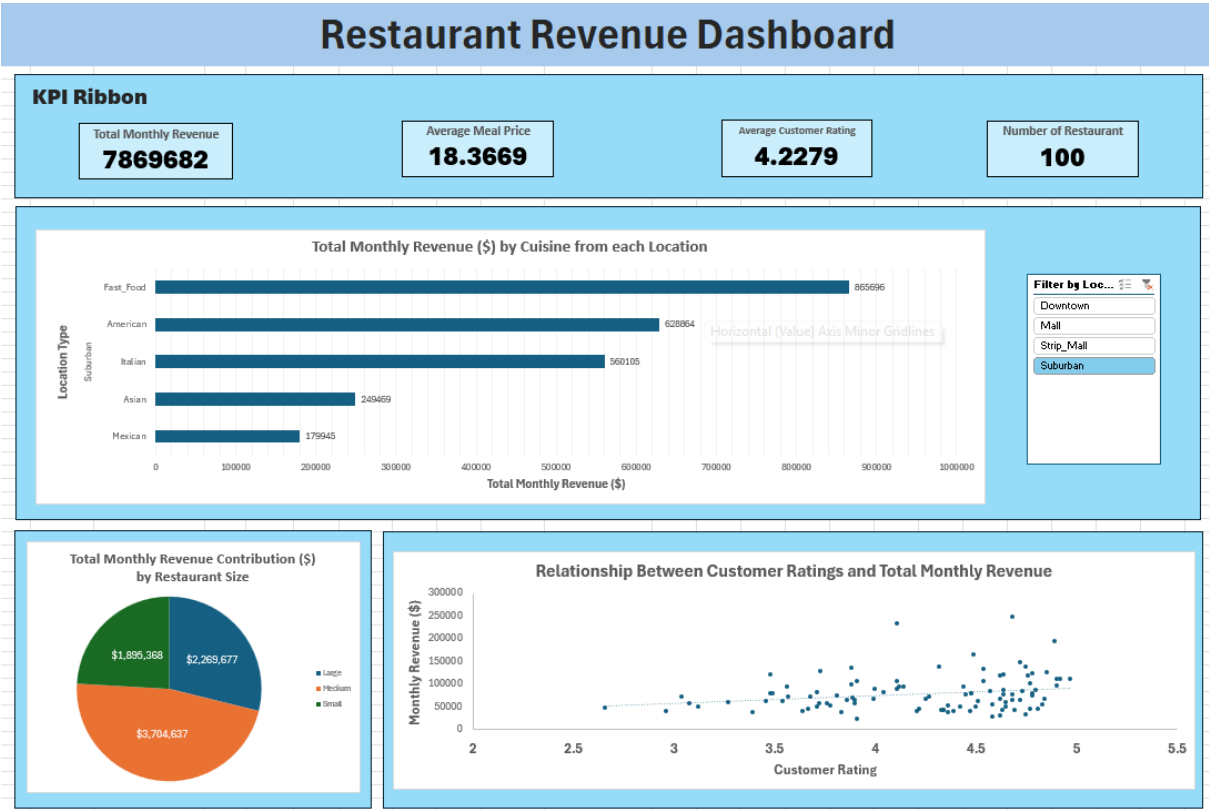
6.0 Conclusion

In conclusion, this study analyzes the relationship between restaurant characteristics and monthly revenue using a Multiple Linear Regression model. Model 1, which includes all quantitative variables, has the highest standard error, with two variables being statistically insignificant to monthly revenue, and it violates the MLR assumption of homoscedasticity, as seen in the residual plot. Model 2 includes all variables available in the dataset except seating_capacity, which was excluded due to a multicollinearity issue. It has a lower standard error and a higher Adjusted R^2 value than Model 1; however, its residuals are also heteroscedastic. Finally, Model 3 is an improved model in which the dependent variable was log-transformed. Several attributes that were statistically insignificant in Model 2 were removed. Residual diagnostics showed randomly distributed residuals, satisfying all regression assumptions. All remaining predictors in Model 3 were statistically significant, demonstrating that each variable contributes meaningfully to explaining restaurant revenue.

The results show that location type plays a crucial role, with restaurants in Mall and Downtown areas generating higher revenue compared to those in Suburban areas. In addition, customer ratings have a positive effect on revenue, underscoring the importance of service quality and customer experience. Other factors, such as marketing spend, average meal price, staff count, and years in business, also positively influence revenue.

Overall, this study demonstrates that restaurant revenue is determined by a combination of strategic location decisions, investment, and customer-focused practices. The findings enable restaurant owners to improve restaurant performance and ultimately increase monthly revenue.

7.0 Appendix



Restaurant Revenue Dashboard

The prototype dashboard was designed using Excel to provide an overview of business operations based on the 100 restaurant records, illustrating how different attributes contribute to revenue. It allows internal teams and potential clients to quickly understand the financial position of a restaurant. The dashboard includes several KPI indicators, such as Total Monthly Revenue, Average Meal Price, Average Customer Rating, and Number of Restaurants, based on the dataset. It provides a general overview of restaurant business operations across the 100 restaurants.

A horizontal bar chart was included to show Total Monthly Revenue by Cuisine Type from each location. This interactive chart allows clients to identify how each location impacts revenue for each cuisine type, helping them identify the best location for expansion. For example, in the dashboard, a restaurant owner opening an Asian restaurant could consider locating in a Strip Mall, as Asian restaurants generated the highest proportion of total monthly revenue in Strip Malls compared to other locations.

A pie chart at the bottom left of the dashboard shows the revenue split between Large, Medium, and Small restaurants. It helps potential clients understand how restaurant scale affects monthly revenue. In the dashboard, medium-sized restaurants contribute the largest share of revenue among the restaurant size categories. This suggests that medium-sized restaurants achieve an optimal balance of seating capacity, operating costs, and number of staff. This insight may help clients with small restaurants consider expanding their restaurant size for greater revenue opportunities.

A scatter plot is included in the dashboard to show the relationship between customer ratings and total monthly revenue. It provides a visual representation of the correlation between customer satisfaction and revenue. Based on the dashboard, a positive relationship is observed between customer rating and monthly revenue, meaning that the higher the customer rating, the higher the monthly revenue generated by restaurants. This helps clients understand how improving a restaurant's rating — through promotional campaigns, better customer service, and consistent food quality — can enhance revenue.

Overall, the dashboard translates data into actionable business insight through visualization, illustrating how each factor affects monthly revenue in alignment with the objectives of restaurant owners.